

Module 4(b) - Active Filters

Active Filters : Ideal Responses, First Order Stages, VCVS Unity Gain Second Order Low pass Filters, VCVS Equal Component Low Pass Filters, VCVS High Pass Filters, MFB Bandpass Filters, Band-stop Filters.

Introduction- Almost all communication systems use filters. A filter passes one band of frequencies while rejecting another. Filters can separate desired signals from undesired signals, block interfering signals, enhance speech and video, and alter signals in other ways.

Q1. What are Active filters? Explain their classification.

Filters are frequency selective circuits that pass only desired band of frequencies and attenuates the undesired band of frequencies.

Classification of Filters

Filters are classified based on the following parameters

Parameter	Remarks
(i) Type of Signal	<u>Analog and Digital filters</u> Analog filters are used to process analog signals, where as Digital filters process analog signals using digital techniques.
(ii) Based on Component/s	<u>Passive and Active Filters</u> Passive filters use only passive components such as Inductors, Capacitors and Resistors and are generally used above 1 MHz, have no power gain, and are relatively difficult to tune. Whereas Active filters use active devices like Transistors or op-amps together with passive components are useful and below 1 MHz, have power gain, and are relatively easy to tune.
(iii) Operating Frequency	<u>AF and RF filters.</u> RC filters are used for audio frequencies (AF) where LC and crystal filters are used at radio frequency (RF).
(iv) Frequency response	Based on the frequency response, filters are classified into (a) Low Pass Filters (LPF) (b) High Pass Filters (HPF) (c) Band Pass Filters (BPF) (d) Band Rejection Filters (BRF) (e) All Pass filters.
(v) In terms of the rate at which the output falls off at the edge of the frequency	→ If the <u>fall off rate</u> is 20dB/decade, then the filter is called as First order filter. → If the <u>roll-off rate</u> is 40 dB/decade, then the filter is Second order filter. → If the <u>gain fall rate</u> is 60 dB/decade, then the filter is called Third order filter and so on.

Note - The frequency response of a filter is the graph of its voltage gain versus frequency.

Q2. List the advantages of Active Filters over Passive Filters

- (a) Because of high input impedance and low output impedance of op-amp, the active filters doesn't cause loading of the source or load. Hence there exists an excellent isolation between source and load.
- (b) Active filters are more compact and economical due to the absence of bulky, costly inductors.
- (c) Due to the availability of modern IC's and cheaper op-amps, active filters are cheap in cost.
- (d) The gain of the active filter can be easily controlled and also it is easier to tune or adjust the active filter to any desired frequency compared to passive filters.

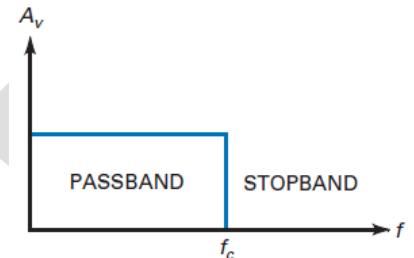
Q3. Explain the ideal response of filters

Figure shows the frequency response characteristics of the various active filters. The ideal response is shown by the dashed lines. The roll-off region between the passband and the stopband is called the **transition**.

Ideal Responses

(i) Lowpass filter

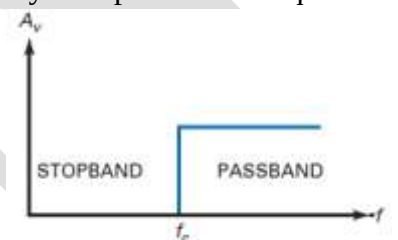
A low-pass filter passes all frequencies from zero to the cutoff frequency and blocks frequencies above the cutoff frequency. The frequencies between zero and the cutoff frequency constitute the Passband and the frequencies above the cutoff frequency are called the Stopband. An ideal low-pass filter has zero attenuation (signal loss) in the passband, infinite attenuation in the stopband, and a vertical transition



(2) High Pass Filter

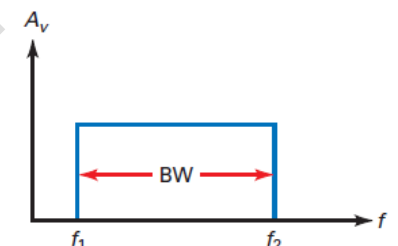
A High-pass filter blocks all frequencies from zero up to the cutoff frequency and passes all frequencies above the cutoff frequency. With a high-pass filter, the frequencies between zero and the cutoff frequency are the Stopband and the frequencies above the cutoff frequency are the Passband.

An ideal high-pass filter has Infinite attenuation in the Stopband, zero attenuation in the Passband, and a vertical transition.



(3) Bandpass filter

A Bandpass filter passes only a specific range of frequencies and all others are blocked. All frequencies from zero up to the lower cutoff frequency are blocked. Then, it passes all the frequencies between the lower and upper cutoff frequencies. Finally, it blocks all frequencies above the upper cutoff frequency. With a bandpass filter, the Passband is all the frequencies between the Lower and Upper cutoff frequencies. The frequencies below the lower cutoff frequency and above the upper cutoff frequency are the Stopband. An ideal bandpass filter has zero attenuation in the passband, infinite attenuation in the stopband, and two vertical transitions.



The center frequency $f_o = \sqrt{f_1 f_2}$ Bandwidth $BW = f_2 - f_1$ and Quality factor $Q = \frac{f_o}{BW}$

If $[Q < 1]$, the bandpass filter is called a **Wideband filter**.

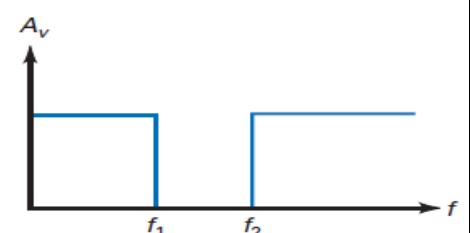
If $[Q > 1]$, the filter is called a **Narrowband filters**. For $Q > 10$, the center frequency $f_o = \left(\frac{f_1 + f_2}{2}\right)$

Ex- FM transmission where the signal frequencies from 88 MHz to 108 MHz has to be transmitted

(4) Bandstop Filter

This type of filter passes all frequencies from zero up to the lower cutoff frequency. Then, it blocks all the frequencies between the lower and upper cutoff frequencies. Finally, it passes all frequencies above the upper cutoff frequency. Hence there exists two passbands and one stopband.

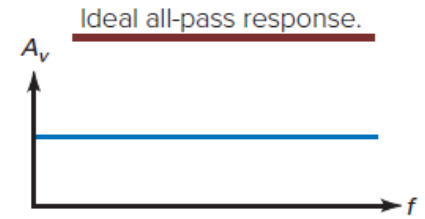
The center frequency $f_o = \sqrt{f_1 f_2}$ Bandwidth $BW = f_2 - f_1$ and Quality factor $Q = \frac{f_o}{BW}$



(5) All pass filter

All pass filter has a passband and with no stopband. Because of this, it passes all frequencies between zero and infinite frequency. The reason it is called a filter is because of the effect it has on the *phase* of signals passing through it. The all-pass filter is useful when it is required to produce a certain amount of phase shift for the signal being filtered without changing its amplitude.

Ex- An opamp inverting amplifier with gain 1



Q4. Explain the practical response of a filter

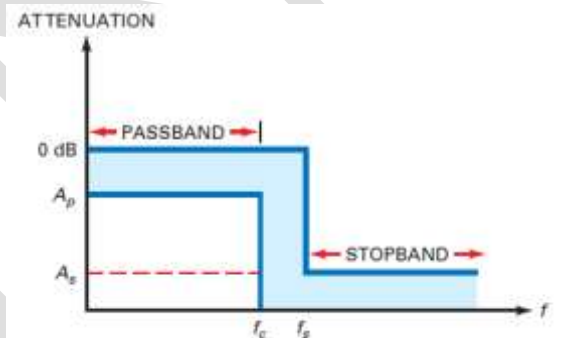
Practical response of the filter

Figure shows the practical response of a low-pass filter, it has three regions – (i) The passband is the set of frequencies between 0 and f_c . (ii) The stopband is all the frequencies above f_s . (iii) The transition region is between f_c and f_s .

The passband is allowing for an attenuation between 0 and A_p instead of zero attenuation. Similarly, the stopband no longer has infinite attenuation, allowing the stopband attenuation to be anywhere from A_s to infinity.

The transition region is no longer vertical, accepting a nonvertical roll-off. The roll-off rate will be determined by the values of f_c , f_s , A_p , and A_s .

Note - The highest frequency in the passband of a low-pass filter is called the cutoff frequency (f_c). This frequency is also referred to as the **edge frequency** because it is on the edge of the passband. In some filters, the attenuation at the edge frequency is less than 3 dB. For this reason, we will use f_{3dB} for the frequency when the attenuation is down 3 dB and f_c for the edge frequency, which may have a different attenuation.



Q5. Define the following with respect to active filters (a) Phase response (ii) Attenuation and (c) Order of the filter

The Phase response of a filter is defined as the graph of phase shift versus frequency. With the all-pass filter, each distinct frequency can be shifted by a certain amount as it passes through the filter.

Attenuation refers to a loss of signal. With a constant input voltage, attenuation is defined as the ratio of output voltage at any frequency to the output voltage in the midband.

$$\text{Attenuation} = \frac{V_{\text{out}}}{V_{\text{out (midband)}}$$

Attenuation is normally expressed in decibels using this equation:

$$\text{Decibel attenuation} = -20 \log_{10} [\text{attenuation}]$$

(For an attenuation of 0.5, the decibel attenuation is: Decibel attenuation = $-20 \log_{10} [0.5] = 6 \text{ dB}$)

Order of Filter (n)

The **Order of a Passive filter** (symbolized by n) equals the number of inductors and capacitors in the filter. If a passive filter has two inductors and two capacitors, $n = 4$. If a passive filter has five inductors and five capacitors, $n = 10$.

The **Order of an Active filter** depends on the number of RC circuits or number of capacitors (called **poles**) it contains. If an active filter contains eight RC circuits, then $n = 8$.

Q6. Why do we need approximations? Compare different filter approximations

The ideal responses are impossible to realize with practical circuits, but there are five standard approximations used as compromises for the ideal responses. Each of these approximations offers an advantage that the others do not have. The approximation chosen by a designer will depend on what is acceptable in an application

Comparison of Filter approximations

Type	Passband	Stopband	Roll-off	Step response
Butterworth	Flat	Monotonic	Good	Good
Chebyshev	Rippled	Monotonic	Very good	Poor
Inverse Chebyshev	Flat	Rippled	Very good	Good
Elliptic	Rippled	Rippled	Best	Poor
Bessel	Flat	Monotonic	Poor	Best

First- Order Stages

First-order or 1-pole active-filter stages have only one capacitor. Because of this, they can produce only a low-pass or a high-pass response. Bandpass and bandstop filters can be implemented only when n is greater than 1.

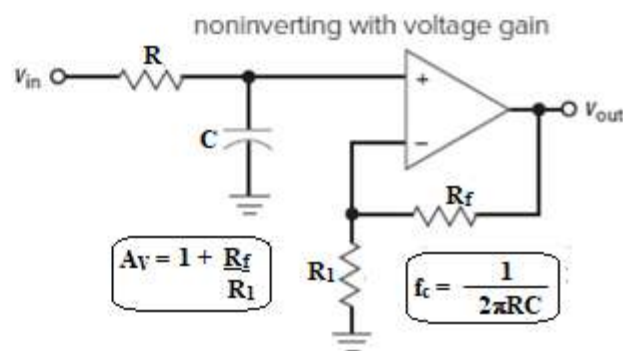
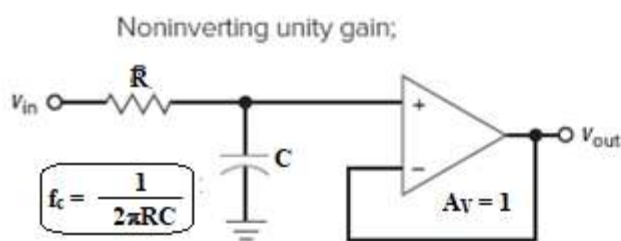
Q7. With a neat circuit diagram, explain first order Lowpass filter

First order Lowpass filter

(a) Non-Inverting Low-Pass Stage

Figure shows the first-order low-pass active filter implemented using (i) an voltage follower with gain $A_V = 1$

(ii) a non inverting amplifier with gain $A_V = \left[1 + \left(\frac{R_2}{R_3}\right)\right]$ & one RC lag circuit



Analysis of the Filter Circuit

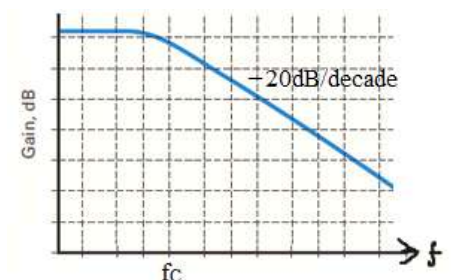
The potential at the non-inverting terminal is

$$V_A = \left[\frac{-jX_{C1}}{R_1 - jX_{C1}} \right] V_{in} = \frac{\left[\frac{1}{j2\pi f C_1} \right] V_{in}}{R_1 + \frac{1}{j2\pi f C_1}} = \frac{V_{in}}{1 + j2\pi f R_1 C_1}$$

The output voltage V_O is given by $V_O = A_V V_A = A_V \left[\frac{V_{in}}{1 + j2\pi f R_1 C_1} \right]$

$$\frac{V_O}{V_{in}} = \left[\frac{A_V}{1 + j\left(\frac{f}{f_c}\right)} \right]$$

Where $f_c = \frac{1}{2\pi R_1 C_1}$ is the high frequency cut – off of the filter



$$\left| \frac{V_o}{V_{in}} \right| = \frac{A_v}{\sqrt{1 + (f/f_c)^2}} \quad \text{and phase angle } \phi = \tan^{-1} \left(\frac{f}{f_c} \right)$$

The operation of the circuit can be summarized as

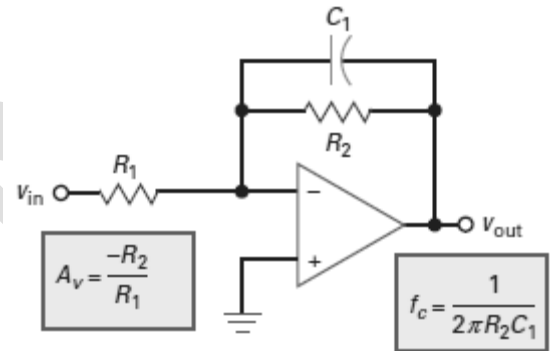
$$(i) \text{ For } f < f_c; \rightarrow \frac{V_o}{V_{in}} \cong A_v \quad (ii) \text{ for } f = f_c; \rightarrow \frac{V_o}{V_{in}} \cong 0.707A_v \quad (iii) f > f_c; \rightarrow \frac{V_o}{V_{in}} < A_v \cong 0$$

Thus in the frequency range; $0 < f < f_c$, the gain is almost constant and is equal to A_v and at $f = f_c$, the gain is $0.707 A_v$, and after f_c the gain decreases at a constant rate of -20dB/decade with increase in frequency. The frequency f_c is called cut-off frequency because the gain of the filter is down by 3 dB from 0 Hz.

Q8. With a neat circuit diagram, explain first order active Low pass stage using inverting amplifier

Figure shows the first-order low-pass active filter implemented using an inverting amplifier gain $A_v = -\left(\frac{R_2}{R_1}\right)$ and one RC lag circuit

The parallel combination of C_1 and R_2 sets the cut-off frequency. At low frequencies the capacitor reactance is much higher than R_2 , so the dc gain $A_v = (-R_2/R_1)$. As the frequency increases the capacitors reactance decreases reducing the impedance of the parallel combination of $X_C \parallel R_2$, in-turn the gain of the amplifier until eventually at a high enough frequency, X_C reduces to zero resulting in zero output.



$$A_v = -\left(\frac{R_2}{R_1}\right) \quad f_c = \frac{1}{2\pi R_2 C_1}$$

The operation of the circuit can be summarized as

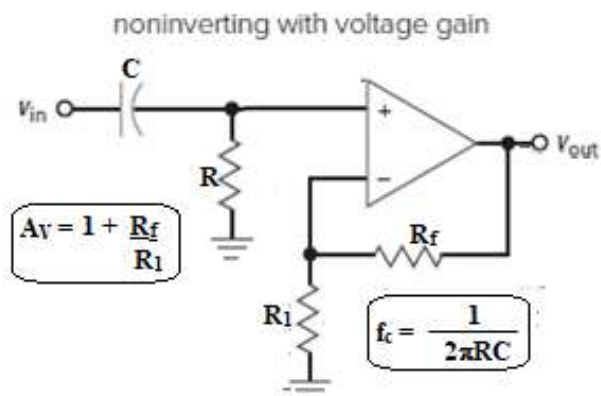
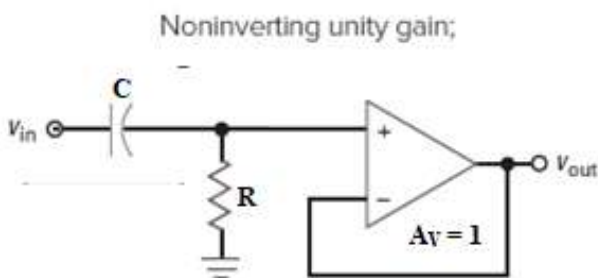
$$(i) \text{ For } f < f_c; \rightarrow \frac{V_o}{V_{in}} \cong A_v \quad (ii) \text{ for } f = f_c; \rightarrow \frac{V_o}{V_{in}} \cong 0.707A_v \quad (iii) f \gg f_c; \rightarrow \frac{V_o}{V_{in}} \cong 0$$

Q9. With a neat circuit diagram, explain first order active high pass filter

(c) Non-Inverting first order High-Pass Stage

Figure shows the first-order low-pass active filter implemented using (i) an voltage follower with gain $A_v = 1$

(ii) a non inverting amplifier with gain $A_v = \left[1 + \left(\frac{R_2}{R_3}\right)\right]$ & one RC lag circuit



Analysis of the Filter Circuit

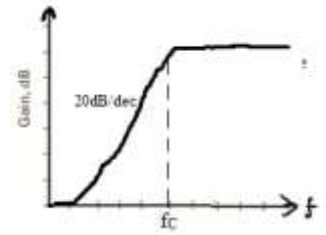
The potential at the non-inverting terminal is

$$V_A = \left[\frac{R_1}{R_1 - jX_{C1}} \right] V_{in} = \frac{R_1 V_{in}}{R_1 + \frac{1}{j2\pi f C_1}} = \left[\frac{j2\pi f R_1 C_1}{1 + j2\pi f R_1 C_1} \right] V_{in}$$

The output voltage V_O is given by

$$V_O = A_V V_A = A_V \cdot \left[\frac{j2\pi f R_1 C_1}{1 + j2\pi f R_1 C_1} \right] V_{in} \quad \text{or} \quad \frac{V_O}{V_{in}} = A_V \left[\frac{j\left(\frac{f}{f_C}\right)}{1 + j\left(\frac{f}{f_C}\right)} \right]$$

where $f_C = \frac{1}{2\pi R_1 C_1}$ is the low frequency cut – off of the filter and $\left| \frac{V_O}{V_{in}} \right| = A_V \left[\frac{j\left(\frac{f}{f_C}\right)}{1 + j\left(\frac{f}{f_C}\right)} \right]$



The operation of the circuit can be summarized as

(i) $f < f_C \rightarrow \frac{V_O}{V_{in}} \cong 0$ (ii) for $f = f_C \rightarrow \frac{V_O}{V_{in}} \cong 0.707 A_V$ (iii) For $f > f_C \rightarrow \frac{V_O}{V_{in}} \cong A_V$

Thus, the circuit acts as high pass filter with a passband gain as A_V . For the frequencies, $f < f_C$, the gain increases till $f = f_C$ at a rate of + 20 dB/decade. Hence, the slope of the frequency response in stop band is + 20 dB/decade for first order high pass filter.

Q10. With a neat circuit diagram, explain first order active high-pass stage using inverting amplifier

High pass stage using inverting amplifier

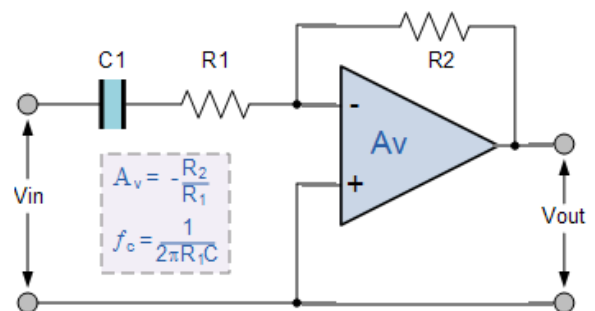
Figure shows the first-order High-pass active filter implemented using an inverting amplifier with gain $A_V = -\left(\frac{R_2}{R_1}\right)$ and one RC lag circuit

The series combination of C_1 and R_1 sets the -3dB frequency. At low frequencies the capacitors reactance is much higher than R_2 , so the dc gain is almost zero. As the frequency increases the capacitors reactance decreases reducing the impedance of the series combination of X_{C1} and R_1 , until the impedance becomes only R_1

$$A_V = -\left(\frac{R_2}{R_1}\right) \quad f_C = \frac{1}{2\pi R_1 C_1}$$

The operation of the circuit can be summarized as

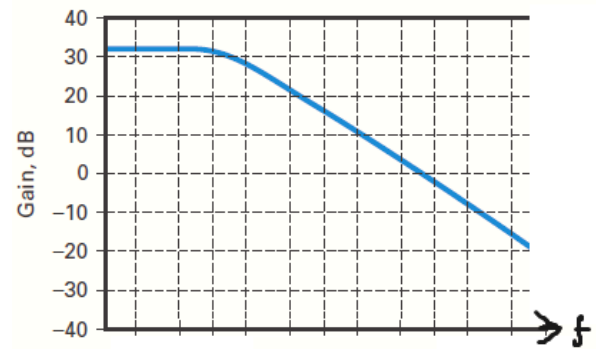
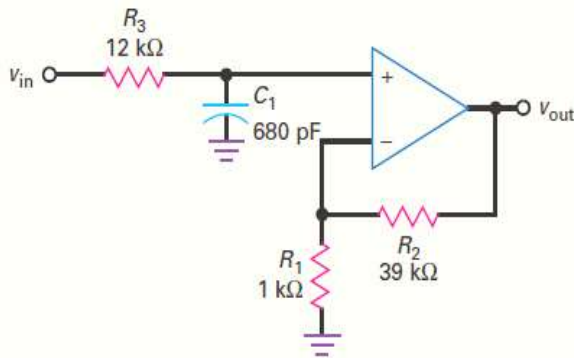
(i) For $f < f_C \rightarrow \frac{V_O}{V_{in}} \cong 0$
(ii) for $f = f_C \rightarrow \frac{V_O}{V_{in}} \cong 0.707 A_V$ (iii) $f > f_C \rightarrow \frac{V_O}{V_{in}} \cong A_V$



Problem – Determine the voltage gain, the cutoff frequency and draw the frequency response for the filter circuit shown below. (ii) If the resistor R_1 is changed to 6.8 k Ω . Find the new cutoff frequency.

Solution - The given circuit is a Lowpass filter using non-inverting amplifier. The gain of the filter is

$$A_V = \left[1 + \left(\frac{R_2}{R_1} \right) \right] = \left(1 + \frac{39}{1} \right) = 40 \quad \text{or} \quad A_{V(\text{dB})} = 20 \log 40 = 32 \text{ dB}$$



$$\text{Cut-off frequency } f_c = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(12\text{k})(680 \times 10^{-12})} = 19.5 \text{ kHz}$$

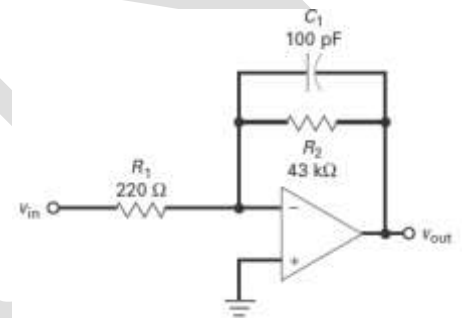
For $R_1 = 6.8 \text{ k}\Omega$, the new cut-off frequency f_{c1} is

$$f_{c1} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(6.8\text{k})(680 \times 10^{-12})} = 34.41 \text{ kHz}$$

Problem – Determine the voltage gain, the cutoff frequency and draw the frequency response for the filter

$$A_v = -\left(\frac{R_2}{R_1}\right) = -\left(\frac{43 \times 10^3}{220}\right) = -135$$

$$f_c = \frac{1}{2\pi R_2 C_1} = \frac{1}{2\pi \times 43 \times 10^3 (100 \times 10^{-12})} = 37 \text{ kHz}$$



Problem – Design a first order non inverting Lowpass filter with a cut-off frequency of 1kHz with a voltage gain of 11.

$$\text{Solution – Given } A_v = 11 = \left[1 + \left(\frac{R_2}{R_1}\right)\right] \quad \text{or} \quad \left(\frac{R_2}{R_1}\right) = 11 - 1 = 10$$

$$\text{Let } R_1 = 1 \text{ k}\Omega, \text{ then } R_2 = 10 \text{ k}\Omega \quad \text{Given } f_c = 1 \text{ kHz} \quad \text{and} \quad f_c = \frac{1}{2\pi R_1 C_1}$$

$$\text{Let } C_1 = 0.01 \mu\text{F}, \text{ then } R_1 = \frac{1}{2\pi f_c C_1} = \frac{1}{2\pi \times 1 \times 10^3 (0.01 \times 10^{-6})} = 15.91 \text{ k}\Omega$$

VCVS Second- Order Filters

Second-order or 2-pole stages are the most common because they are easy to build and analyze. Higher-order filters are usually made by cascading second order stages. Each second-order stage has a resonant frequency and a Q to determine how much peaking occurs.

These filters are also called as **Sallen-Key filters**. These filters are also called VCVS filters because the op amp is used as a voltage-controlled voltage source. VCVS low-pass circuits can implement three of the basic approximations: Butterworth, Chebyshev, and Bessel.

Filter type	Peaked Responses			Remarks
Lowpass filter	$f_c = K_C f_p$	$f_0 = K_O f_p$	$f_{3dB} = K_3 f_p$	Butterworth filter $Q = 0.707$; $K_C = K_3 = 1$ and $K_O = \text{NA}$
Highpass filter	$f_c = (f_p/K_C)$	$f_0 = (f_p/K_O)$	$f_{3dB} = (f_p/K_3)$	Bessel Filter $Q = 0.577$; $K_C = 0.786$ $K_3 = 1$ and $K_O = \text{NA}$

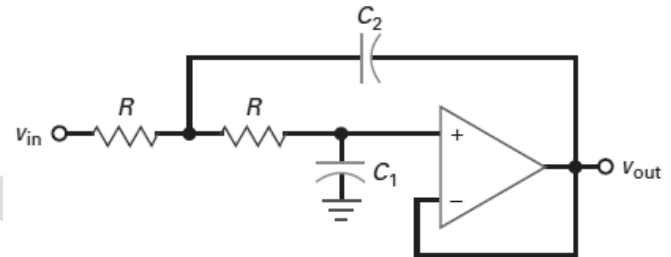
Q11. Explain the operation of VCVS (i) Unity-Gain Second- Order Lowpass Filter (ii) Equal-Component Second order Low-Pass Filter (Second order Lowpass filter with gain)

VCVS Unity-Gain Second- Order Lowpass Filter

Figure shows a Sallen-Key second-order low-pass filter. The two resistors have the same value, but the two capacitors are different. There is a lag circuit on the non-inverting input, but this time, there is a feedback path through a second capacitor C_2 .

Operation - At low frequencies, both capacitors appear to be open and the circuit has a unity gain because the op amp is connected as a voltage follower.

As the frequency increases, the impedance of C_1 and C_2 decreases and the gain eventually starts to decrease until it is down 3dB at the cut-off frequency and gain roll-off takes place at the rate -40dB/ decade . As the frequency increases above cut-off frequency, the output is attenuated.



The pole frequency is given by $f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}}$ and the Quality factor $Q = 0.5 \sqrt{\frac{C_2}{C_1}}$

For Butterworth filter; $K_C = 1$

Cut-off or Edge frequency $f_C = K_C f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}}$ Hz

For Bessel filter; $K_C = 0.786$

Cut-off or Edge frequency $f_C = K_C f_p = \frac{0.786}{2\pi R \sqrt{C_1 C_2}}$ Hz

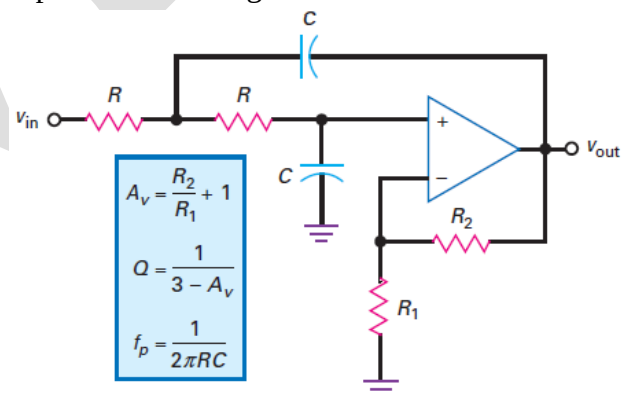
(ii) VCVS Equal-Component Second order Low-Pass Filter (Second order Lowpass filter with gain)

Figure shows a Sallen-Key second-order equal component low-pass filter with gain. Both resistors and both capacitors have the same value.

Operation - At low frequencies, both capacitors appear to be open and the circuit has a gain because the op amp is connected as a non-inverting amplifier with gain.

$$A_V = \left[1 + \left(\frac{R_2}{R_1} \right) \right]$$

As the frequency increases, the impedance of C_1 and C_2 decreases and the gain eventually starts to decrease until it is down 3dB at the cut-off frequency and gain roll-off takes place at the rate -40dB/ decade . As the frequency increases above cut-off frequency, the output is attenuated.



$$\begin{aligned} A_V &= \frac{R_2}{R_1} + 1 \\ Q &= \frac{1}{3 - A_V} \\ f_p &= \frac{1}{2\pi RC} \end{aligned}$$

The pole frequency is given by $f_p = \frac{1}{2\pi RC}$ and the Quality factor $Q = \left(\frac{1}{3 - A_V} \right)$

As A_V increase from 1 to 3, the Q value increases from 0.5 to ∞ . Hence the range for the gain is $1 \leq A_V < 3$. For $A_V \geq 3$, the circuit goes into oscillations because of larger feedback and gain.

For Butterworth filter; $K_C = 1$

Cut-off or Edge frequency $f_C = K_C f_p = \frac{1}{2\pi RC}$ Hz

For Bessel filter; $K_C = 0.786$

Cut-off or Edge frequency $f_C = K_C f_p = \frac{0.786}{2\pi RC}$ Hz

Q12. Explain the operation of VCVS (i) Unity Gain Second- Order Highpass Filter (ii) Equal-Component Second order Highpass Filter (Second order Lowpass filter with gain)

(i) VCVS Unity-Gain Second- Order Highpass Filter

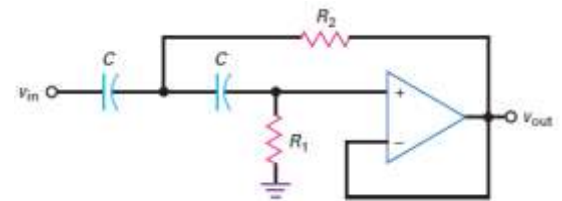
Figure shows a Sallen-Key second-order Highpass filter. The two Capacitors have the same value, but the two resistors are different. There is a lag circuit on the non-inverting input, but this time, there is a feedback path through a second capacitor C_2 .

Operation - At low frequencies, both capacitors appear to be open and no input is applied to the non inverting terminal, hence output is zero.

As the frequency increases, the impedance of C_1 and C_2 decreases and the gain eventually starts to increase and becomes unity after cutoff frequency. The gain roll-on takes place at the rate +40dB/ decade. As the frequency increases above cut-off frequency, the output is constant.

The pole frequency is given by $f_p = \frac{1}{2\pi C \sqrt{R_1 R_2}}$ and

the Quality factor $Q = 0.5 \sqrt{\frac{R_1}{R_2}}$



For Butterworth filter; $K_C = 1$

Cut-off or Edge frequency $f_c = \frac{f_p}{K_C} = \frac{1}{2\pi C \sqrt{R_1 R_2}}$ Hz

For Bessel filter; $K_C = 0.786$

Cut-off or Edge frequency $f_c = \frac{f_p}{K_C} = \frac{1}{2\pi C \sqrt{R_1 R_2 (0.786)}}$ Hz

(ii) VCVS Equal-Component Second order Highpass Filter (Second order Highpass filter with gain)

Figure shows a Sallen-Key second-order equal component high pass filter with gain. Both resistors and both capacitors have the same value.

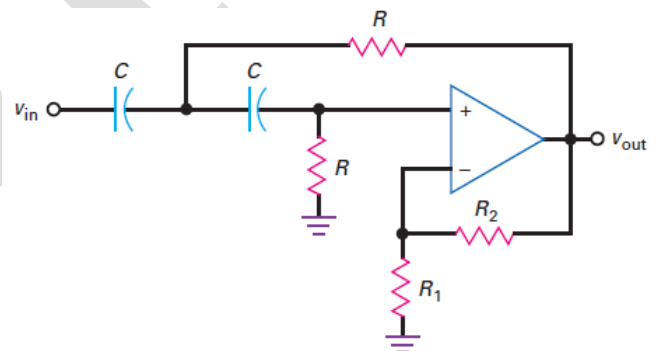
Operation - At low frequencies, both capacitors appear to be open and the circuit and no input is applied, hence the output is zero.

As the frequency increases, the impedance of C_1 and C_2 decreases and the gain eventually starts to increase and becomes $A_V = \left[1 + \left(\frac{R_2}{R_1}\right)\right]$ after cutoff frequency. The gain roll-on takes place at the rate +40dB/ decade. As the frequency increases above cut-off frequency, the output is constant with a gain A_V .

The pole frequency is given by $f_p = \frac{1}{2\pi RC}$ and the

Quality factor $Q = \left(\frac{1}{3 - A_V}\right)$

As A_V increase from 1 to 3, the Q value increases from 0.5 to ∞ . Hence the range for the gain is $1 \leq A_V < 3$. For $A_V \geq 3$, the circuit goes into oscillations because of larger feedback and gain.



For Butterworth filter; $K_C = 1$

Cut-off or Edge frequency $f_c = (f_p / K_C) = \frac{1}{2\pi RC}$ Hz

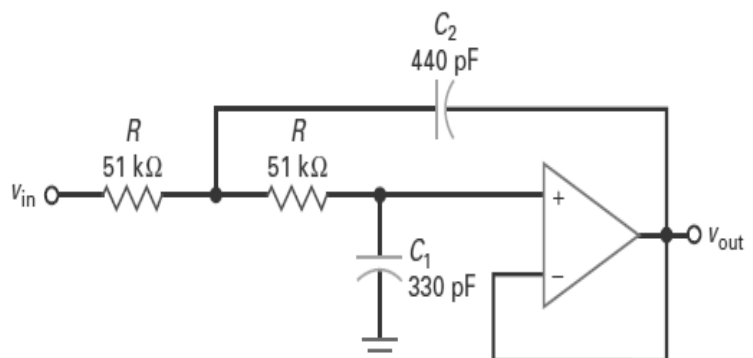
For Bessel filter; $K_C = 0.786$

Cut-off or Edge frequency $f_c = (f_p / K_C) = \frac{1}{2\pi RC (0.786)}$ Hz

Problem - What are the pole frequency and Q in the circuit shown . What is the, cutoff frequency of the filter ?

Solution - The given filter is a II order equal component unity gain lowpass filter

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{440}{330}} = 0.577$$



Since $Q = 0.577$, it is a Bessel filter, then $K_C = 0.786$

$$\text{Pole frequency } f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi \times 51 \times 10^3 \sqrt{330 \times 10^{-9} \times 440 \times 10^{-9}}} = 8.19 \text{ kHz}$$

$$\text{Cut-off frequency } f_c = K_C f_p = (0.786)(8.19) = 6.14 \text{ kHz}$$

Problem - What are the pole frequency and Q in the circuit shown . What is the, cutoff frequency of the filter

Solution - The given filter is a II order equal component lowpass filter

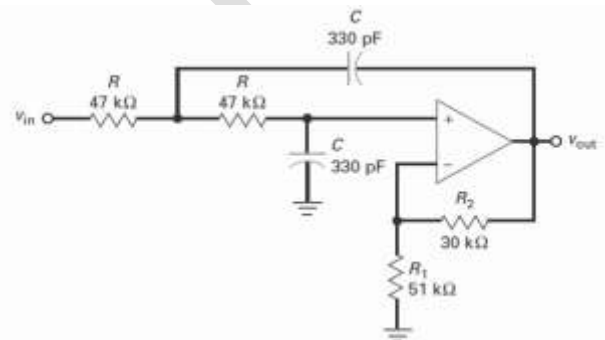
$$\text{Gain } A_V = \left[1 + \left(\frac{R_2}{R_1} \right) \right] = \left[1 + \left(\frac{30}{51} \right) \right] = 1.59$$

$$\text{Quality factor } Q = \left(\frac{1}{3 - A_V} \right) = \left(\frac{1}{3 - 1.59} \right) = 0.709$$

Since $Q = 0.709 \cong 0.707$, the filter is approximated as Butterworth filter, hence $f_p = f_c$

$$f_p = f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 47 \times 10^3 \times 330 \times 10^{-9}}$$

$$f_p = f_c = 10.3 \text{ kHz}$$



Problem - What are the pole frequency and Q in the circuit shown . What is the, cutoff frequency of the filter ?

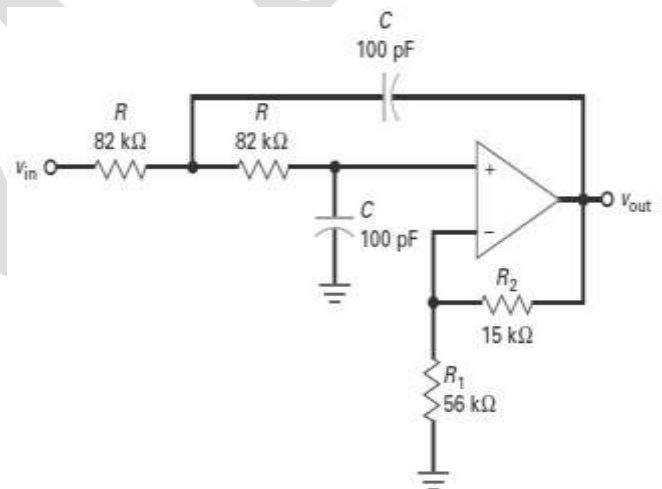
Solution - The given filter is a II order equal component lowpass filter

$$\text{Gain } A_V = \left[1 + \left(\frac{R_2}{R_1} \right) \right] = \left[1 + \left(\frac{15}{56} \right) \right] = 1.27$$

$$\text{Quality factor } Q = \left(\frac{1}{3 - A_V} \right) = \left(\frac{1}{3 - 1.27} \right) = 0.578$$

Since $Q = 0.578$, the filter is approximated as Bessel filter, hence $f_c = K_C f_p$ and for Bessel Filter $K_C = 0.786$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 82 \times 10^3 \times 100 \times 10^{-12}} = 19.4 \text{ kHz}$$



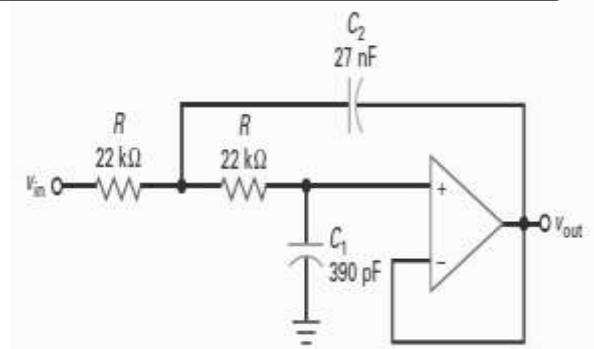
What are the pole frequency and Q in the circuit shown below. What are the resonant, cutoff, and 3-dB frequencies? Given $K_O = 0.99$, $K_C = 1.38$ and $K_3 = 1.54$

Solution - The given filter is a II order equal component unity gain lowpass filter

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{27 \times 10^{-9}}{390 \times 10^{-12}}} = 4.16$$

$$\text{Pole frequency } f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}}$$

$$f_p = \frac{1}{2\pi \times 22 \times 10^3 \sqrt{390 \times 10^{-12} \times 27 \times 10^{-9}}} = 2.23 \text{ kHz}$$



Cut-off frequency $f_c = K_C f_p = (1.38)(2.23) = 3.08 \text{ kHz}$

Resonant frequency $f_o = K_{Op} = 0.99(2.23) = 2.23 \text{ kHz}$

3dB frequency $f_{3dB} = K_3 f_p = 1.54(2.23) = 3.43 \text{ kHz}$

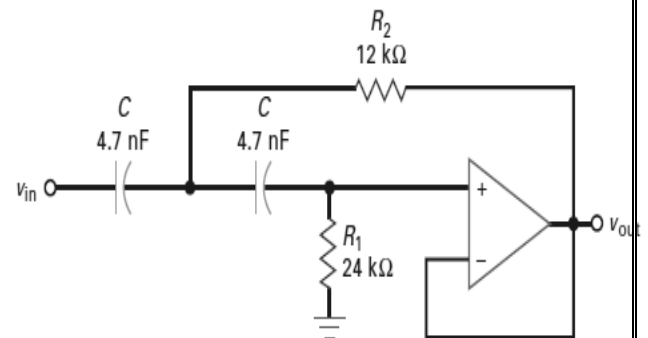
Problem - What are the pole frequency and Q in the circuit shown . What is the, cutoff frequency of the filter ?

Solution - The given filter is a II order equal component unity gain Highpass filter

$$Q = 0.5 \sqrt{\frac{R_1}{R_2}} = 0.5 \sqrt{\frac{24}{12}} = 0.707$$

Since $Q = 0.707$, it is a Butterworth filter

$$f_c = f_p = \frac{1}{2\pi C \sqrt{R_1 R_2}} = \frac{1}{2\pi \times 4.7 \times 10^{-9} \sqrt{24 \times 10^3 \times 12 \times 10^3}} = 2 \text{ kHz}$$



What are the pole frequency and Q in the circuit shown below. What are the resonant, cutoff, and 3-dB frequencies? Given $K_O = 0.94$, $K_C = 1.32$ and $K_3 = 1.48$.

Solution - The given filter is a II order equal component Highpass filter

$$A_V = \left[1 + \left(\frac{R_2}{R_1} \right) \right] = \left[1 + \left(\frac{15}{10} \right) \right] = 2.5$$

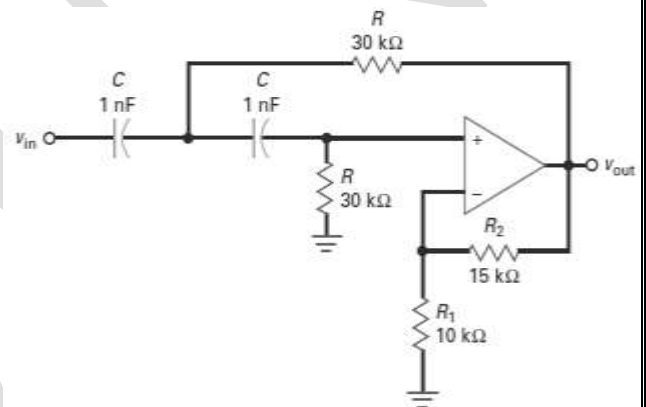
$$Q = \left(\frac{1}{3 - A_V} \right) = \left(\frac{1}{3 - 2.5} \right) = 2$$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi (30 \times 10^3)(1 \times 10^{-9})} = 5.31 \text{ kHz}$$

Cut-off frequency $f_c = (f_p / K_C) = (5.31 / 1.32) = 4.02 \text{ kHz}$

Resonant frequency $f_o = (f_p / K_o) = (5.31 / 0.94) = 5.65 \text{ kHz}$

3dB frequency $f_{3dB} = (f_p / K_3) = (5.31 / 1.48) = 3.59 \text{ kHz}$



Problem Design a II order Butterworth Lowpass filter for a cut of frequency of 3

Solution - For Butterworth filter $Q = 0.707$

$$Q = 0.707 = \left(\frac{1}{3 - A_V} \right) \text{ or } A_V = 1.59$$

$$A_V = 1.59 = \left[1 + \left(\frac{R_2}{R_1} \right) \right] \text{ or } \frac{R_2}{R_1} = 0.59$$

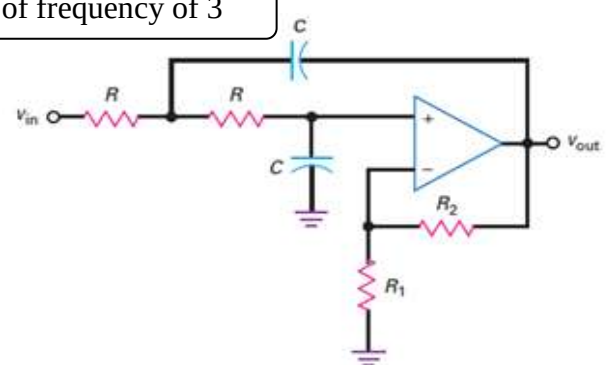
Let $R_1 = 10 \text{ k}\Omega$, then $R_2 = 5.9 \text{ k}\Omega$

For Butterworth filter

Pole frequency $f_p =$ Cut-off frequency $f_c = 3 \text{ kHz}$

Let $C = 0.01 \text{ }\mu\text{F}$

$$f_c = \frac{1}{2\pi RC} \text{ or } R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi (3 \times 10^3)(1 \times 10^{-9})} = 5.31 \text{ k}\Omega$$



Problem Design a II order Bessel Lowpass filter for a cut of frequency of 3 kHz

Solution - For Bessel filter $Q = 0.587$

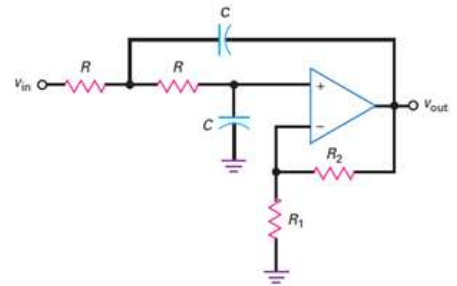
$$Q = 0.587 = \left(\frac{1}{3 - A_V} \right) \text{ or } A_V = 1.3$$

$$A_V = 1.3 = \left[1 + \left(\frac{R_2}{R_1} \right) \right] \text{ or } \frac{R_2}{R_1} = 0.3$$

Let $R_1 = 10 \text{ k}\Omega$, then $R_2 = 3 \text{ k}\Omega$

Given cut-off frequency $f_c = 3000 \text{ Hz}$

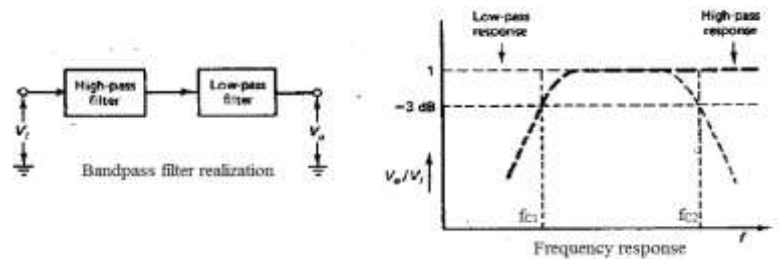
$$f_c = \frac{1}{2\pi RC} \quad \text{or} \quad R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi(3 \times 10^3)(1 \times 10^{-9})} = 5.31 \text{ k}\Omega$$



Multiple Feedback (MFB) Bandpass Filters

Q13. Write an explanatory note on Multiple Feedback (MFB) Bandpass Filters

A Band Pass Filter Circuit designed to pass signals only in a certain band of frequencies while rejecting all signals outside this band. There are basically two types of Band Pass Filter Circuit,



(i) Wide band pass (ii) Narrow band pass

A Bandpass filter is defined as a Wide bandpass if its figure of merit or quality factor $Q < 1$ and if $Q > 1$, then filter is referred to as Narrow Bandpass filter. Hence Q is a measure of selectivity meaning, the higher the value of Q , the more selective is the filter, or the narrower is the band width. The relationship between Q , 3dB bandwidth and the centre frequency f_0 is given by

$$(i) BW = f_{c2} - f_{c1} \quad (ii) Q = \frac{f_c}{BW} = \frac{f_c}{f_{c2} - f_{c1}} \quad \text{and} \quad (iii) f_0 = \sqrt{f_{c1} f_{c2}}$$

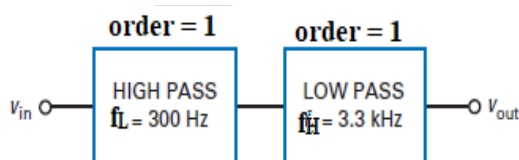
Where f_{c2} and f_{c1} are cut-off frequencies of Lowpass and Highpass filters.

Figure shows the frequency response of a band pass filter. This type of filter has a maximum output voltage $V_{\max}$ at one frequency called the resonant frequency, f_0 . If the frequency is varied away from resonance, the output voltage decreases. There is one frequency above f_0 and one below f_0 at which the voltage is $0.707V_{\max}$ (3dB point). These frequencies are the high and low cutoff frequencies. The band of frequencies between f_H and f_L is the bandwidth. Therefore the band width is given by $BW = f_{c2} - f_{c1}$

Q14. Explain the working of Wide Bandpass filter.

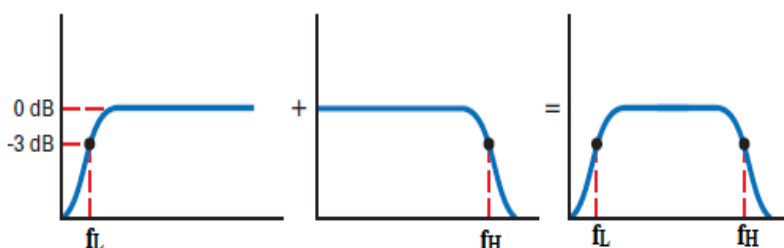
Wide Bandpass Filter

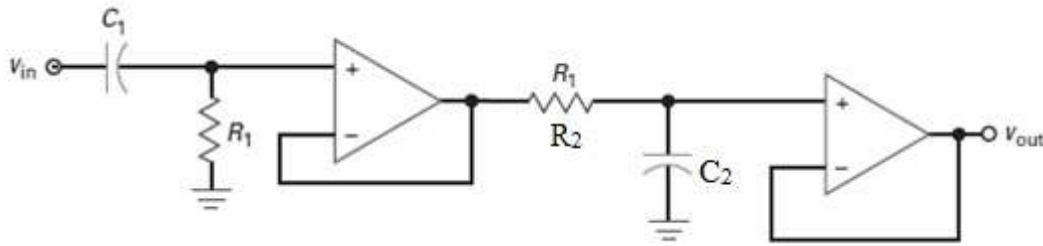
A wide bandpass filter ($Q < 1$) can be formed by simply cascading high pass and low pass section and is generally the choice for simple to design. To obtain a $\pm 20 \text{ dB/decade}$ band pass filter, a first order high pass



filter and a first low pass sections are cascaded, for a $\pm 40 \text{ dB/decade}$ band pass filter,

Figure shows the block diagram of a $\pm 20 \text{ dB/decade}$ wide band pass filter (which is composed of a first order high pass and a first order low pass filter) and its frequency response. Consider the cascaded Highpass and Lowpass filters as shown in the block diagram





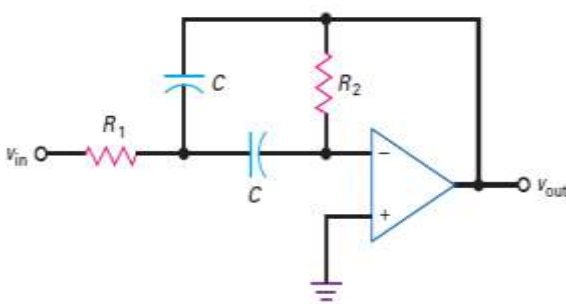
Ex – (i) $BW = f_{c2} - f_{c1} = 3300 - 300 = 3\text{kHz}$ (ii) $f_o = \sqrt{f_{c1}f_{c2}} = \sqrt{(3.3)(0.3)} = 995\text{ Hz}$

(iii) $Q = \frac{f_o}{BW} = \frac{995}{3000} = 0.332$

Q15. With a neat circuit diagram and necessary equations explain the operation of Narrow bandpass filter. Discuss how the input impedance of the stage can be increased.

Narrow bandpass Filters

When Q is greater than 1, the bandpass filter is known as Narrow bandpass filter and can be implemented using multiple feedback. The input signal goes to the inverting input first rather than the non-inverting input and the circuit has two feedback paths, one through a capacitor and another through a resistor.



At low frequencies, the capacitors appear to be open. Therefore, the input signal cannot reach the op amp, and the output is zero. At high frequencies, the capacitors appear to be shorted. In this case, the voltage gain is zero because the feedback capacitor has zero impedance. Between the low and high extremes in frequency, there is a band of frequencies where the circuit acts like an inverting amplifier.

The voltage gain at the center frequency is given by:

$$A_V = -\left(\frac{R_2}{2R_1}\right)$$

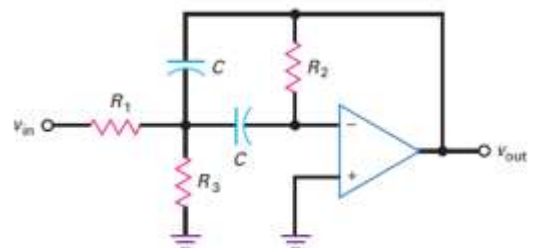
Center frequency $f_o = \frac{1}{2\pi C\sqrt{R_1R_2}}$ and the Quality factor $Q = 0.5\sqrt{\frac{R_2}{R_1}} = 0.5\sqrt{-2A_V} = 0.707\sqrt{-A_V}$

Increasing input impedance of MFB stage.

Since the above circuit uses inverting amplifier whose input impedance is low, to increase the input impedance, the modified circuit is used. The modified equations are

$$A_V = -\left(\frac{R_2}{2R_1}\right) \quad f_o = \frac{1}{2\pi C\sqrt{(R_1||R_3)R_2}}$$

and $Q = 0.5\sqrt{\frac{R_2}{R_1||R_3}}$



Bandstop Filters (Notch Filters)

Q16. With a neat circuit diagram and necessary equations, explain the operation Bandstop filter

A bandstop filter needs to block only a single frequency. Figure shows a Sallen-Key second-order Notch filter

Operation At low frequencies, all capacitors are open. As a result, all the input signal reaches the non-inverting input. The circuit has a passband voltage gain of: $A_V = \left[1 + \left(\frac{R_2}{R_1}\right)\right]$

At very high frequencies, the capacitors are shorted. Again, all the input signal reaches the non-inverting input. The circuit has a passband

voltage gain of: $A_V = \left[1 + \left(\frac{R_2}{R_1}\right)\right]$

Between the low and high extremes in frequency, there is a center frequency given by:

$$f_0 = \frac{1}{2\pi RC}$$

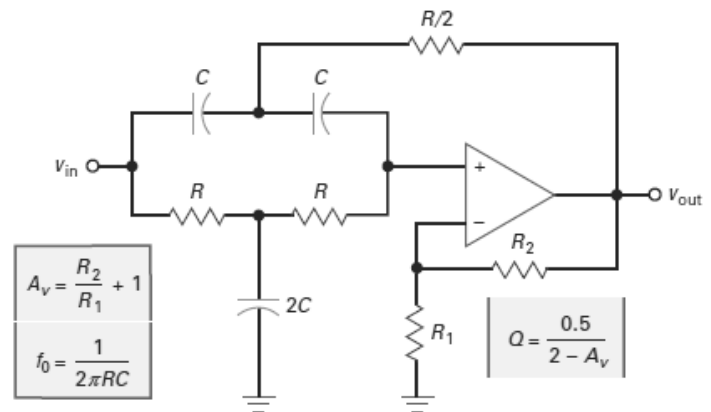
At this frequency, the feedback signal returns with the correct amplitude and phase to attenuate the signal on the non-inverting input. Because of this, the output voltage drops to a very low value.

The Q of the circuit is given by: $Q = \left[\frac{0.5}{2 - A_V}\right]$

To avoid oscillations and proper operation of the circuit (i) the voltage gain A_V has to be less than 2 and (ii) Q should be much less than 10.

Ex- The ac power lines may induce a hum of 60 Hz in sensitive circuits; this may interfere with a desired signal. A bandstop filter is used to notch out the unwanted hum signal.

Sallen-Key second-order notch filter.



Problem – (a) Determine the voltage gain, the center frequency and Q for the bandstop filter with $R_1 = 13\text{k}\Omega$, $R_2 = 10\text{k}\Omega$, $R = 22\text{k}\Omega$ and $C = 120\text{nF}$. (b) change R_2 to obtain a Q value of 3. Also, change the C value for a center frequency of 120 Hz.

Solution

(a) Gain $A_V = \left[1 + \left(\frac{R_2}{R_1}\right)\right] = \left[1 + \left(\frac{10}{13}\right)\right] = 1.77$

Center frequency $f_0 = \frac{1}{2\pi RC} = \frac{1}{2\pi(22\text{k}\Omega)(120\text{nF})} = 60.3 \text{ Hz}$

Quality factor $Q = \left[\frac{0.5}{2 - A_V}\right] = \left[\frac{0.5}{2 - 1.77}\right] = 2.17$

(b) For $Q = 3 \Rightarrow 3 = \left[\frac{0.5}{2 - A_V}\right]$ or $A_V = \left[\frac{6 - 0.5}{3}\right] = 1.833$

For Gain $A_V = 1.833$; $1.833 = \left[1 + \left(\frac{R_2}{R_1}\right)\right]$ or $R_2 = [1.833 - 1]R_1 = 0.833(13) = 10.83 \text{ k}\Omega$

New Center frequency $f_0 = 120 \text{ Hz}$ hence $C = \frac{1}{2\pi R f_0} = \frac{1}{2\pi(22\text{k}\Omega)(120)} = 60.3 \text{ }\mu\text{F}$